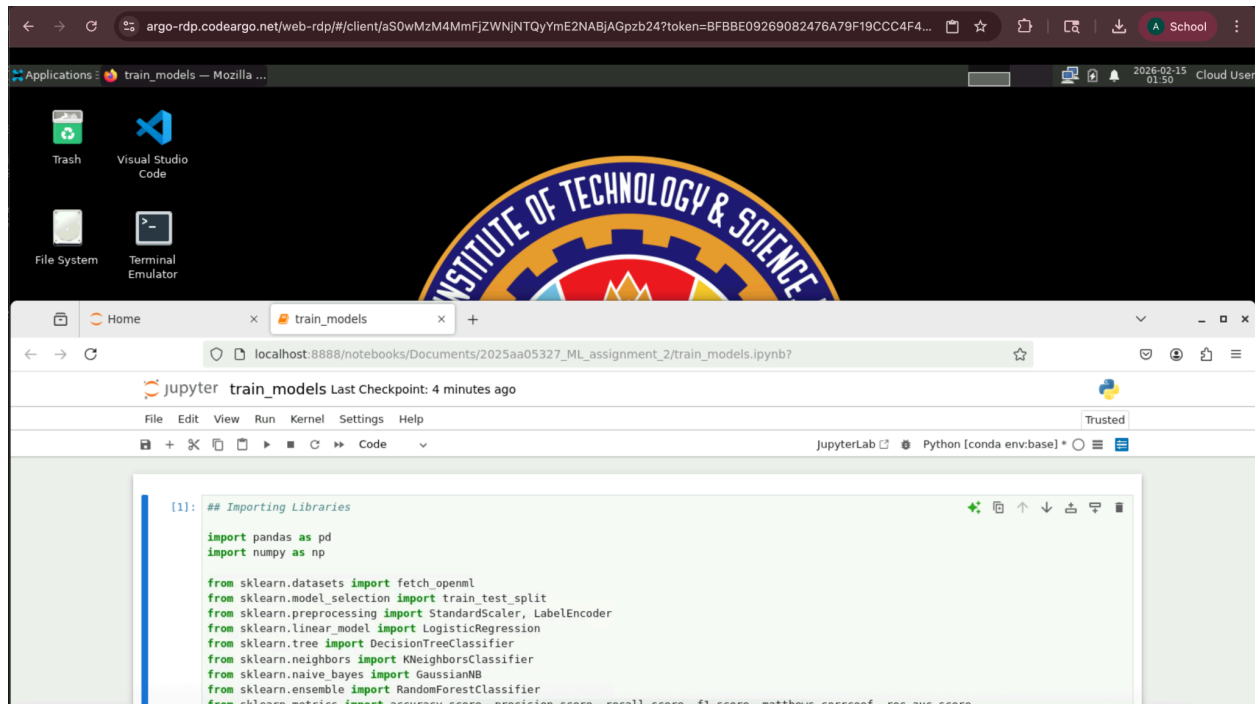


1. GitHub Repository Link - https://github.com/anirbansen2709/2025AA05327_ML_Assignment2
2. Live Streamlit App Link - <https://2025aa05327mlassignment2.streamlit.app/>
3. Screenshot -



4. The Github README content -

Machine Learning Assignment 2

1. Problem Statement

The goal of this project is to implement and evaluate six different classification algorithms to predict Default of Credit Card Clients using the UCI dataset (<https://archive.ics.uci.edu/dataset/350/default+of+credit+card+clients>). The project involves data preprocessing, model training, evaluation using multiple metrics, and deployment of the final solution as an interactive Streamlit web application.

2. Dataset Description

- Dataset Name: Default of Credit Card Clients dataset
- Source: <https://archive.ics.uci.edu/dataset/350/default+of+credit+card+clients>

- Description: The dataset consists of 15000 instances and 23 features. It includes attributes such as LIMIT_BAL, EDUCATION etc to classify samples into default yes/no categories.

3. Models Used & Evaluation Metrics

The following six machine learning models were implemented and evaluated. The performance metrics for each model are summarized below:

Model	Accuracy	AUC Score	Precision	Recall	F1-Score	MCC
Logistic_Regres sion	0.8117	0.7113	0.6907	0.2489	0.3659	0.3341
Decision_Tree	0.8187	0.7398	0.6504	0.3664	0.4688	0.3917
KNN	0.7923	0.6883	0.5385	0.3420	0.4183	0.3109
Naive_Bayes	0.3900	0.6670	0.2453	0.8641	0.3822	0.1193
Random_Forest	0.8033	0.7271	0.5881	0.3313	0.4238	0.3352
XGBoost	0.8073	0.7363	0.5932	0.3740	0.4588	0.3626

4. Observations on Model Performance

Below are the specific observations regarding how each model performed on the chosen dataset:

ML Model Name	Observation about model performance	
Logistic Regression	Strong Baseline: Achieved a solid 81.1% accuracy. While reliable for non-defaults, it had the lowest recall for defaults (0.25), indicating it struggles to identify risky clients compared to non-linear models.	
Decision Tree	Top Performer: Surprising leader in this run with the highest Accuracy (0.818) and AUC (0.740). It achieved a better balance for the minority class (Recall 0.37) than the linear baseline.	
KNN Classifier	Moderate: Performance was lower than tree-based models (AUC 0.688). This suggests that "closeness" in the feature space is not the strongest indicator of credit risk in this specific 23-feature dataset.	
Naive Bayes	Poor Fit: The model failed significantly on accuracy (0.39) because it over-predicted defaults. While it had high recall (0.86), the extremely low precision makes it impractical for this use case.	
Random Forest (Ensemble)	Under-utilized: Performed slightly worse than the single Decision Tree. This is likely because it was restricted to only 10 estimators, which wasn't enough to leverage the power of the ensemble.	
XGBoost (Ensemble)	Robust & Competitive: Delivered a strong AUC of 0.736. It matched the Decision Tree's recall (0.37) for defaults, showing it is effectively handling the complex relationships in the data.	

5. Project Links

- Live App: <https://2025aa05327mlassignment2.streamlit.app/>
- GitHub Repository: https://github.com/anirbansen2709/2025AA05327_ML_Assignment2

6. How to Run Locally

1. Clone the repository:

```
git clone  
https://github.com/anirbansen2709/2025AA05327_ML_Assignment2.git
```

2. Install dependencies:

```
pip install -r requirements.txt
```

3. Run the Streamlit app:

```
streamlit run app.py
```